SIDDHANTA MOHANTY

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EDUCATION

The Pennsylvania State University — University Park, PA
Bachelor of Science in Computer Science — *Aug. 2022 – May 2026*

- **Dean's List:** Fall 2023, Spring 2024, Fall 2025, Spring 2026 **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Artificial Intelligence, Computer Organization, Theory of Computation, Deep Learning for Computer Vision

EXPERIENCE

Backend Developer (Volunteer) — Oct. 2026 – Present
Mentor A Promise — *Remote*

- Build **REST APIs** and database models for a **Learning Management System (LMS)** using Django, supporting course management, instructor workflows, and role-based access control for NYC shelter students.
- Develop **frontend interfaces** using **React and TypeScript**, translating **Figma designs** into instructor dashboards and course settings interfaces for course creation and management.

Research Assistant — Sep. 2025 – Dec. 2025
Penn State University — *University Park, PA*

- Evaluated **reinforcement learning (RL)** approaches to improve code correction in an existing two-LLM pipeline that generates, verifies, and iteratively corrects code using verification feedback.
- Proposed **Proximal Policy Optimization (PPO)** and designed a reward framework using **GCC** compiler diagnostics, **KLEE** symbolic execution, and **CodeQL** static analysis results to guide the learning process.

GenAI Intern — May 2025 – Aug. 2025
HCL Technologies — *Redmond, WA*

- Built a **RAG-based support automation system** using Flask, LangChain, and FAISS that analyzed production error logs, retrieved similar incidents and docs, and sent Slack alerts with remediation recommendations, reducing support resolution time by **50%**
- Designed **retrieval pipelines** using FAISS and structured metadata to index internal docs and historical incidents
- Orchestrated **multi-agent workflows** across APIs, databases, and Confluence using **MCP** and LangChain

PROJECTS

Self-Heal System | *Flask, LangChain, FAISS, PostgreSQL, OpenAI, Slack* — Source Code

- Built an **AI-powered self-healing backend** that embedded error logs into FAISS, retrieved similar past incidents via semantic search, and generated context-aware remediation recommendations.
- Developed a **Slack-integrated incident response system** that delivered severity-based alerts, LLM-generated diagnostics and fixes, and a chatbot with access to recent error logs, database statistics, and a read-only SQL interface restricted to **SELECT** queries.

Rewind | *Python, Playwright, Pydantic, Claude API, MCP* — Source Code

- Built an **LLM-driven browser agent** using Claude tool-calling and Playwright that completes multi-step workflows on a demo banking app from accessibility-tree snapshots and compiles successful runs into **typed, versioned replay artifacts**.
- Engineered a **deterministic replay engine** with **zero LLM calls** that runs recorded workflows (e.g., loan requests) in **0.41s vs 13.1s** (**32x faster**), exposed to AI agents through an **MCP server** with allowlisting, approval gating, and secret redaction.

Prompt Tracer | *JavaScript, Chrome Extension API, Manifest V3* — Source Code

- Developed a **prompt observability Chrome extension** that analyzes and scores prompts in real time using a deterministic 5-point rubric, injecting a live feedback panel into ChatGPT, Claude, Grok, and Gemini for instant, consistent scoring and suggestions
- Built a dual-mode prompt optimization engine combining a privacy-safe rule-based optimizer with an optional OpenAI-powered rewrite pipeline, using ambiguity detection to distinguish prompts requiring clarification from those needing structural improvements

yourAIbrief | *Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, Resend, Vercel* — Demo | Source Code

- Built a full-stack **AI newsletter platform** with magic-link authentication, personalized topic preferences and delivery schedules, aggregating content from 50+ trusted sources and tracking per-issue email open and click analytics.

TECHNICAL SKILLS

Languages: Python, C#, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks/Libraries: Flask, .NET, FastAPI (REST APIs), Django, Next.js, Node.js, React, LangChain, SQLAlchemy, Pydantic
AI/ML Tools: RAG, LLMs, AI Agents, Tool Calling, Model Context Protocol (MCP), Claude API, OpenAI API, NumPy, PyTorch
Databases: PostgreSQL, MySQL, MongoDB, Vector Databases (FAISS, Pinecone)
Developer Tools: Git, GitHub, GitHub Actions, Docker, Playwright, Prisma, Postman, Tableau, AWS, Google Cloud Platform, VS Code, ROS2